

Brindlewythe Biosciences Ltd, a Tarnovell company

AP02 second-generation safety panel with CYP3A4 concentration response

Final Report

Study BW21_0453 (LIMS B-ACC-2021-07710)

Client: Avelnelwick Pharmaceuticals Ltd

Study lead: Dr L. Benmezroun, Study Director

Report date: 25 November 2021

1 Summary

42 compounds from the Avelnelwick aminopyrimidine regioisomer series 02 were tested for Avelnelwick Pharmaceuticals Ltd in the hERG automated patch clamp, 5-isoform CYP panel, 3A4 concentration response. Testing started on 08 November 2021 and results were reported on 25 November 2021. All accepted runs met plate acceptance criteria; one run was rejected and repeated (see Section 6). No compound had a hERG IC₅₀ below 1 µM. The set divided almost evenly around 10 µM (20 below, 22 at or above 10 µM or above the top concentration). Inhibition above 50% at 10 µM was most frequent for CYP3A4; CYP1A2 was rarely affected.

Table 1 Study summary

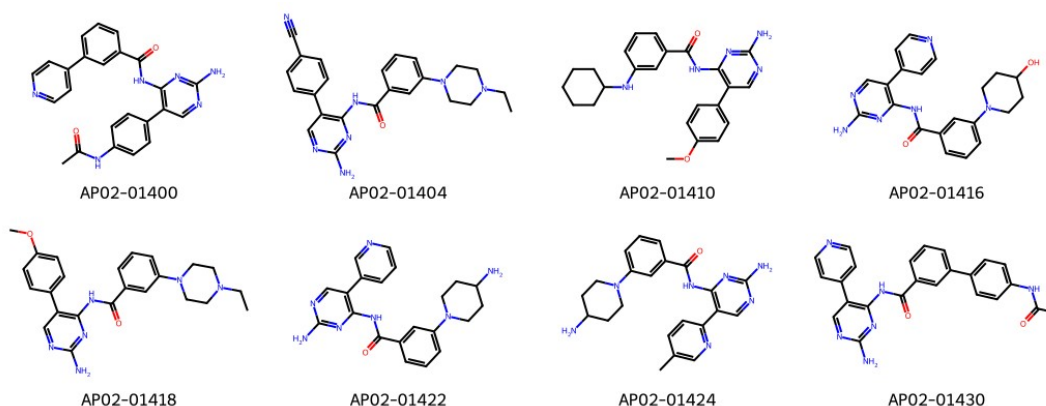
Metric	Value
Project	BW21_0453
Client	Avelnelwick Pharmaceuticals Ltd
Compounds tested	42
Series / chemotype	Avelnelwick aminopyrimidine regioisomer series 02 (Aminopyrimidine regioisomer)
Assays	hERG patch clamp, CYP inhibition panel (1A2/2C9/2C19/2D6/3A4) at 10 µM, CYP3A4 inhibition IC ₅₀
Turnaround (calendar days)	17
Price	GBP 15,900
hERG IC ₅₀ <1 µM	0
hERG IC ₅₀ 1-10 µM	20
hERG IC ₅₀ ≥10 µM	22
hERG median IC ₅₀ (µM)	10
E-4031 control IC ₅₀ (nM)	14
CYP1A2 >50% inhibition @10 µM	1
CYP2C9 >50% inhibition @10 µM	3
CYP2C19 >50% inhibition @10 µM	11
CYP2D6 >50% inhibition @10 µM	7
CYP3A4 >50% inhibition @10 µM	13
Compounds with any CYP >50%	17
Ketoconazole (3A4) IC ₅₀ (µM)	0.029

2 Objective

To rank the screening set for hERG automated patch clamp, 5-isoform CYP panel, 3A4 concentration response liability so that Avernell Pharmaceuticals Ltd could choose analogues for progression.

3 Test items

42 compounds were received as 10 mM DMSO solutions in one test lot on 08 November 2021 and logged against B-ACC-2021-07710. Identity and concentration were taken as supplied by the client. Stocks were stored at -20 °C and thawed once for testing.



4 Methods

hERG automated patch clamp

HEK293 cells stably expressing hERG (Kv11.1) were recorded in whole-cell configuration on a 16-channel automated patch clamp platform at room temperature. Cells are held at -80 mV, depolarised to +40 mV for 2 s and repolarised to -50 mV for 2 s to elicit the tail current, at 0.1 Hz. Each compound was applied cumulatively at 0.12, 0.37, 1.1, 3.3, 10 and 30 μ M (0.3% DMSO final). Each concentration is held until the tail-current block reaches a steady state (change below 5% over the last three sweeps), with a minimum of 3 and a maximum of 8 minutes; compounds still drifting at 8 minutes are flagged as slow-onset.

E-4031 is run as a separate 6-point concentration response (1-300 nM) on every run.

- Plate acceptance: seal resistance ≥ 500 M Ω before break-in and ≥ 200 M Ω throughout
- Tail current ≥ 200 pA at baseline; run-down $\leq 10\%$ over the vehicle period
- E-4031 IC₅₀ within 5-50 nM; runs outside this range are rejected and repeated
- IC₅₀ fitted per cell (four-parameter logistic); the compound IC₅₀ is the geometric mean of finite cell IC₅₀s when these are the majority of accepted cells, otherwise reported as >30 μ M
- Where precipitation is seen at 30 μ M, that concentration is excluded and the highest evaluable concentration is reported

5-isoform CYP panel

Pooled human liver microsomes (0.1 mg/mL protein) were incubated with each compound at 10 μ M and an isoform-selective probe substrate at approximately K_m, in duplicate, for 10 minutes at 37 °C after NADPH addition. Metabolite formation is measured by LC-MS/MS against vehicle (0.1% DMSO)

incubations; the lower limit of quantitation of each metabolite is at least 20-fold below the vehicle signal.

Probe substrates and controls

Isoform	Probe substrate	Metabolite	Positive control
CYP1A2	Phenacetin	Acetaminophen	Furafylline
CYP2C9	Diclofenac	4'-Hydroxydiclofenac	Sulfaphenazole
CYP2C19	S-Mephenytoin	4'-Hydroxymephenytoin	Ticlopidine
CYP2D6	Dextromethorphan	Dextrorphan	Quinidine
CYP3A4	Midazolam	1'-Hydroxymidazolam	Ketoconazole

Positive controls are run as 7-point concentration responses on every run. Furafylline and ticlopidine (mechanism-based inhibitors) are preincubated with microsomes and NADPH for 15 minutes before probe addition. A run is accepted when each control IC₅₀ falls within 0.5-2 times the laboratory historical mean. Results are reported as mean percent inhibition relative to vehicle; values above 50% at 10 μ M are flagged, and small negative values indicate no inhibition.

3A4 concentration response

All compounds were run in a 7-point CYP3A4 concentration response (top concentration 50 μ M, 3-fold dilutions) using midazolam 1'-hydroxylation as the probe, with ketoconazole on every plate. A single probe is used so that IC₅₀s are comparable across the set; substrate-dependent differences are outside the scope.

5 Results

hERG results

All 42 compounds were tested. No compound had a hERG IC₅₀ below 1 μ M. The set divided almost evenly around 10 μ M (20 below, 22 at or above 10 μ M or above the top concentration). The median IC₅₀ was 10 μ M (0 below 1 μ M, 20 between 1 and 10 μ M, 22 at or above 10 μ M, of which 3 above the highest evaluable concentration). E-4031 gave an IC₅₀ of 14 nM in the accepted runs.

Compounds with QC notes: AP02-01586: First run rejected (E-4031 outside range); repeated. Values for these compounds should be read with the note.

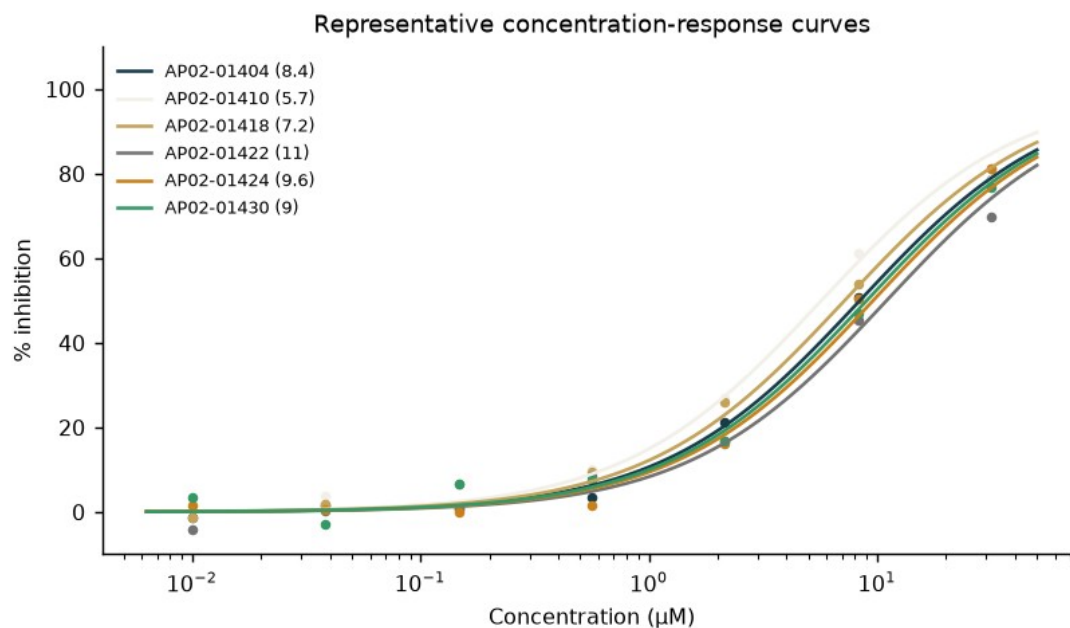
Table 2 hERG IC₅₀ by compound (μ M)

Compound	IC ₅₀ cell 1 (μ M)	IC ₅₀ cell 2 (μ M)	IC ₅₀ cell 3 (μ M)	IC ₅₀ geomean (μ M)	% inhibition @10 μ M	n cells (finite)	QC note
AP02-01400	>30	>30		>30	23	2 (0)	
AP02-01404	8.4	19		13	47	2 (2)	
AP02-01410	5.7	7.2		6.4	56	2 (2)	
AP02-01416	>30	21		>30	22	2 (1)	
AP02-01418	7.2	11	12	9.8	48	3 (3)	
AP02-01422	11	15		13	42	2 (2)	
AP02-01424	9.6	19	11	13	46	3 (3)	
AP02-01430	9.0	12		10	59	2 (2)	
AP02-01434	20	15		18	37	2 (2)	
AP02-01442	24	21		23	30	2 (2)	
AP02-01446	11	8.9		9.9	53	2 (2)	
AP02-01448	4.0	4.3		4.1	69	2 (2)	
AP02-01454	3.7	5.4		4.5	71	2 (2)	

AP02-01458	6.9	5.0		5.8	64	2 (2)	
AP02-01464	10	8.5	8.0	8.9	50	3 (3)	
AP02-01472	21	14	15	16	42	3 (3)	
AP02-01476	11	6.6		8.4	61	2 (2)	
AP02-01484	18	19		18	35	2 (2)	
AP02-01488	4.6	4.2		4.4	75	2 (2)	
AP02-01494	3.1	3.9		3.5	81	2 (2)	
AP02-01496	14	26		19	40	2 (2)	
AP02-01500	5.8	8.7		7.1	63	2 (2)	
AP02-01506	28	>30	>30	>30	26	3 (1)	
AP02-01508	12	11	13	12	47	3 (3)	
AP02-01514	12	17		14	37	2 (2)	
AP02-01518	2.8	2.2		2.5	84	2 (2)	
AP02-01524	29	22		25	28	2 (2)	
AP02-01526	29	18		23	29	2 (2)	
AP02-01532	4.7	3.4		4.0	68	2 (2)	
AP02-01536	3.5	6.2		4.7	71	2 (2)	
AP02-01538	17	19		18	34	2 (2)	
AP02-01544	16	24		20	32	2 (2)	
AP02-01550	11	9.6		10	46	2 (2)	
AP02-01556	9.7	9.5		9.6	49	2 (2)	
AP02-01560	6.7	11		8.7	57	2 (2)	
AP02-01568	10	7.8		8.9	57	2 (2)	
AP02-01572	7.6	9.3		8.4	48	2 (2)	
AP02-01578	13	12	10	12	46	3 (3)	
AP02-01580	5.5	6.5		5.9	62	2 (2)	
AP02-01584	14	17	12	14	42	3 (3)	
AP02-01586	11	8.6	7.1	8.8	55	3 (3)	First run rejected (E-4031 outside range); repeated
AP02-01590	15	13		14	41	2 (2)	

Top concentration 30 μ M; 6-point, 3-fold dilution; positive control E-4031 IC₅₀ 14 nM (acceptance 5-50 nM)

Compound IC₅₀ = geometric mean of finite cell IC₅₀s when they are the majority of accepted cells; otherwise reported as >30 μ M (or >10 μ M where solubility-limited)



CYP inhibition results

Inhibition above 50% at 10 µM was most frequent for CYP3A4; CYP1A2 was rarely affected. In total 17 compounds inhibited at least one isoform by more than 50% at 10 µM (CYP3A4: 13; CYP2D6: 7; CYP2C19: 11; CYP2C9: 3; CYP1A2: 1). All control IC₅₀s were within 0.5-2 times the historical mean (ketoconazole 0.029 µM against a historical mean of 0.030 µM).

Table 3 Percent inhibition at 10 µM

Compound	CYP1A2 (%)	CYP2C9 (%)	CYP2C19 (%)	CYP2D6 (%)	CYP3A4 (%)
AP02-01400	21	13	45	14	65
AP02-01404	-2	-6	2	-8	9
AP02-01410	11	36	85	99	84
AP02-01416	6	-4	3	-7	-2
AP02-01418	2	0	15	5	3
AP02-01422	-1	6	0	-4	5
AP02-01424	-0	3	2	0	3
AP02-01430	21	5	39	13	49
AP02-01434	9	10	26	10	85
AP02-01442	-3	32	65	-2	24
AP02-01446	16	82	83	14	91
AP02-01448	1	4	-0	13	67
AP02-01454	1	9	30	5	21
AP02-01458	14	35	67	45	22
AP02-01464	-5	-1	0	-2	-0
AP02-01472	-1	5	3	16	40
AP02-01476	2	23	52	58	24
AP02-01484	-5	8	23	16	1
AP02-01488	3	1	25	4	35
AP02-01494	64	11	62	16	62
AP02-01496	-4	-2	-6	5	-1
AP02-01500	1	11	0	12	3
AP02-01506	7	0	-8	3	-0
AP02-01508	6	37	11	94	67
AP02-01514	-0	-3	44	2	-0
AP02-01518	5	4	92	100	66
AP02-01524	32	3	13	12	3
AP02-01526	-2	1	-1	7	6
AP02-01532	-0	12	-6	64	20

AP02-01536	-0	-1	78	68	55
AP02-01538	3	-1	3	35	-5
AP02-01544	43	6	64	46	68
AP02-01550	7	3	-2	8	-4
AP02-01556	4	70	89	39	70
AP02-01560	-0	2	-3	-0	-4
AP02-01568	-4	11	12	28	33
AP02-01572	0	5	2	18	11
AP02-01578	48	10	45	33	89
AP02-01580	6	53	76	53	67
AP02-01584	14	-1	-3	14	26
AP02-01586	-7	-1	24	6	3
AP02-01590	0	6	39	14	35

Single concentration 10 μ M, duplicate incubations, mean reported; negative values = no inhibition

Controls (IC₅₀, μ M): furafylline (CYP1A2) 1.9; sulfaphenazole (CYP2C9) 0.33; ticlopidine (CYP2C19) 1.6; quinidine (CYP2D6) 0.055; ketoconazole (CYP3A4) 0.029

Table 4 Positive control IC₅₀s

Control	Isoform	IC ₅₀ (μ M)	Historical mean (μ M)
Furafylline	CYP1A2	1.9	1.8
Sulfaphenazole	CYP2C9	0.33	0.35
Ticlopidine	CYP2C19	1.6	1.2
Quinidine	CYP2D6	0.055	0.060
Ketoconazole	CYP3A4	0.029	0.030

CYP3A4 concentration response

CYP3A4 IC₅₀s were determined for all 42 compounds: 13 had IC₅₀ below 10 μ M, including 1 below 1 μ M. These potent inhibitors are the main CYP liability in the set.

Table 5 CYP3A4 IC₅₀ (μ M)

Compound	IC ₅₀ (μ M)	Probe substrate
AP02-01400	7.9	midazolam
AP02-01404	>50	midazolam
AP02-01410	1.6	midazolam
AP02-01416	>50	midazolam
AP02-01418	>50	midazolam
AP02-01422	>50	midazolam
AP02-01424	>50	midazolam
AP02-01430	12	midazolam
AP02-01434	1.6	midazolam
AP02-01442	23	midazolam
AP02-01446	0.71	midazolam
AP02-01448	3.7	midazolam
AP02-01454	>50	midazolam
AP02-01458	>50	midazolam
AP02-01464	>50	midazolam
AP02-01472	10	midazolam
AP02-01476	37	midazolam
AP02-01484	>50	midazolam
AP02-01488	35	midazolam
AP02-01494	4.1	midazolam
AP02-01496	>50	midazolam
AP02-01500	>50	midazolam
AP02-01506	>50	midazolam
AP02-01508	4.1	midazolam
AP02-01514	>50	midazolam

AP02-01518	7.6	midazolam
AP02-01524	>50	midazolam
AP02-01526	>50	midazolam
AP02-01532	41	midazolam
AP02-01536	9.6	midazolam
AP02-01538	>50	midazolam
AP02-01544	6.7	midazolam
AP02-01550	>50	midazolam
AP02-01556	6.3	midazolam
AP02-01560	>50	midazolam
AP02-01568	43	midazolam
AP02-01572	36	midazolam
AP02-01578	1.1	midazolam
AP02-01580	4.8	midazolam
AP02-01584	20	midazolam
AP02-01586	>50	midazolam
AP02-01590	32	midazolam

7-point curve, top 50 μ M; ketoconazole control run alongside on every plate

6 Deviations

The first hERG run for one compound was rejected because E-4031 fell outside its acceptance range; the compound was retested in a later run and only accepted data are reported.

7 Discussion

No compound had a hERG IC₅₀ below 1 μ M. The set divided almost evenly around 10 μ M (20 below, 22 at or above 10 μ M or above the top concentration). Inhibition above 50% at 10 μ M was most frequent for CYP3A4; CYP1A2 was rarely affected. These are screening data; mechanism statements are outside the scope of this study.

8 Conclusion

Recommended next steps: prioritise compounds with hERG IC₅₀ at or above 10 μ M; for any compound below 10 μ M, compare the IC₅₀ with projected free plasma exposure before progression; repeat the flagged hERG measurements if a selection decision depends on them; deprioritise compounds with CYP3A4 IC₅₀ below 1 μ M and review the below-10 μ M group with the chemistry team.

9 Records

Raw data, run sheets and instrument files are retained for 5 years under B-ACC-2021-07710.

Name	Role	Signature	Date
Leila Benmezroun	Study Director	<u>L Benmezroun</u>	25 November 2021
Ines Alvarado	Assay Quality Coordinator	<u>I Alvarado</u>	24 November 2021